VARUN JOSE MADANU

Machine Learning Engineer | Computer Vision · Deep Learning · 3D / Spatial Media

varunjosemadanu@gmail.com | +15185964160 | linkedin.com/in/varun-jose-madanu | varunjose.github.io/portfolio

SUMMARY

Machine Learning Engineer with a deep-learning and 3D computer-vision research foundation and 5+ years developing, optimizing, and deploying ML models and infrastructure at scale. Designed, trained, and evaluated CNN architectures on 3D facial depth-map data in PyTorch for medical image classification hands-on depth-based 3D visual understanding. Expert in Python and PyTorch, with strengths in large-scale model training and serving, inference optimization, rigorous experimentation, and taking models from research to production. Fluent in leveraging LLM-based tools and AI coding assistants to accelerate prototyping and development. M.S. in Computer Science.

TECHNICAL SKILLS

Machine Learning & Deep Learning: PyTorch (expert), CNNs, neural network design, model training & evaluation, data augmentation, cross-validation, fine-tuning, RLHF, model optimization, rigorous experimentation

Computer Vision & 3D: Depth estimation, 3D depth-map data, image classification, representation learning, embeddings & semantic similarity, computer-vision research

Generative AI: Transformer models, large language models (OpenAI, Anthropic, Hugging Face), generative model evaluation, prompt and output analysis

Languages: Python (expert), C, C++, JavaScript, SQL, Bash

ML Systems & Infrastructure: Distributed training, model serving, inference optimization, MLOps, Docker, Kubernetes, AWS, GCP, Terraform, CI/CD, high-performance computing

AI-Assisted Development: LLM-based coding assistants and AI developer tools for accelerated prototyping, research, and problem-solving

PROFESSIONAL EXPERIENCE

AI Models Evaluation Analyst Mar 2026 – Present

Handshake *United States*

- Run rigorous comparative experiments evaluating ML model output quality and correctness across technical domains, benchmarking competing models and quantifying where one approach improves quality over another.
- Diagnose systematic model failure patterns across evaluation batches and author detailed analyses that inform model-improvement and training cycles.
- Leverage LLM-based tools and AI coding assistants daily to accelerate analysis, prototyping, and problem-solving workflows.

Software Engineer, AI Infrastructure May 2025 – Feb 2026

Excelerate *United States*

- Designed, served, and optimized deep-learning and LLM models on containerized, distributed infrastructure handling 10K+ daily interactions; cut inference latency 55% through parallelism and caching model optimization for deployment.
- Built large-scale ML infrastructure with Docker, Kubernetes, Terraform, and CI/CD on AWS/GCP, supporting distributed training and scalable model serving.
- Integrated state-of-the-art Transformer models with vector search and retrieval pipelines; implemented safety guardrails and PII/PHI-compliant deployment patterns.

Graduate Research Assistant — Deep Learning & 3D Computer Vision Oct 2023 – May 2024

University at Albany (SUNY) *Albany, NY*

- Researched deep learning for 3D computer vision: designed, trained, and evaluated CNN architectures on 3D facial depth-map data for binary medical classification, achieving high accuracy on a non-invasive diagnostics task.
- Implemented and tuned neural network architectures in Python/PyTorch, applying data augmentation and cross-validation to address class imbalance in clinical data.
- Contributed to an academic research paper submission validating ML for depth-based 3D visual diagnostics; mentored undergraduates in Python and deep learning.

Programming Analyst (ML & Data) Jun 2024 – May 2025
New York State Department of Transportation *Albany, NY*

- Developed predictive ML models and large-scale data pipelines in Python, integrating model outputs into production decision systems and improving query throughput 40%.
- Designed and maintained cloud-hosted ML/data infrastructure on GCP at 99.9% availability under government data standards.

Python Developer (ML / NLP) Mar 2022 – Jun 2023
Apollo Home Healthcare Ltd.

- Built NLP-based document classification and extraction models with data-processing pipelines, reducing manual processing time 60%.
- Optimized backend services for systems serving 50K+ records with strict data-privacy compliance.

Associate Software Engineer (ML) May 2020 – Mar 2022
Cognier Insights Pvt. Ltd. *Hyderabad, India*

- Developed and deployed predictive ML models (scikit-learn, XGBoost) and served predictions through production APIs and pipelines for enterprise clients.

SELECTED PROJECTS

3D Depth-Map CNN for Medical Classification (Research)

- Trained CNNs on 3D facial depth maps for obstructive sleep apnea prediction depth-based 3D visual understanding, full train/evaluate loop in PyTorch with augmentation and cross-validation.

Real-Time Video / Streaming System (C, UDP/RTP)

- Built a distributed real-time streaming system with custom packetization, reassembly, and network simulation in low-level C performance-critical systems engineering applicable to spatial-video processing pipelines.

Semantic Retrieval & Embedding Pipeline

- Engineered an end-to-end embedding and vector-search pipeline over a large corpus, applying representation learning and similarity search for ranked retrieval.

EDUCATION

M.S., Computer Science — University at Albany (SUNY) Aug 2023 – May 2025

B.Tech., Computer Science — JNTUH Mar 2018 – Jun 2022